# The Alchemist's Archive

STUDENT ASSIGNMENT

## 01 Chemical Reactions and Equations

Name _____ Batch _____ Date _____

Use the space below each question. Label diagrams and show your working.

**01** CBSE 2023 · 31/1/1 · Q2 — EXPLAIN

Lead nitrate is heated. Identify the brown fumes and write a balanced equation for the reaction.

**02** CBSE 2023 · 31/1/1 · Q3 — SHOW WORKING

Balance $MnO_2 + HCl \rightarrow MnCl_2 + H_2O + Cl_2$. Explain how you chose the water and acid coefficients.

**03** CBSE 2023 · 31/1/1 · Q17 — EXPLAIN

A strip of lead is left in blue copper sulfate solution. Identify the metal deposited and explain the fading blue colour.

**04** CBSE 2023 · 31/1/1 · Q27 — QUICK RESPONSE

In $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$, name the reducing agent and identify its product.

**05** CBSE 2023 · 31/1/1 · Q27 EXPLAIN

For $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$, identify the species oxidised and the species acting as oxidising agent.

**06** CBSE 2024 · 31/1/1 · Q2 EXPLAIN

A white silver chloride sample darkens in sunlight. Name the product responsible and classify the change.

**07** CBSE 2024 · 31/1/1 · Q21 EXPLAIN

Water is added to quicklime. Write the equation, name the main reaction type and predict the temperature change.

**08** CBSE 2024 · 31/1/1 · Q27 SHOW WORKING

Give one balanced precipitation reaction and identify the visible evidence that a new solid has formed.

EXTRA WORKING SPACE

**09** CBSE 2024 · 31/1/1 · Q27 EXPLAIN

An iron nail in copper sulfate develops a reddish coating. Give the equation and describe the change in the solution.

**10** CBSE 2024 · 31/1/1 · Q7 EXPLAIN

In the reaction of $MnO_2$ with hydrochloric acid, state what happens to the oxidation state of manganese and chloride.

**11** CBSE 2025 · 31/1/1 · Q1 SHOW WORKING

Water is electrolysed completely. If 36 g of water decomposes, find the ideal masses of both gases.

**12** CBSE 2025 · 31/1/1 · Q1 SHOW WORKING

The oxygen collected from water electrolysis occupies 18 mL. Find the hydrogen volume at the same conditions and identify its electrode.

EXTRA WORKING SPACE

**13** CBSE 2025 · 31/1/1 · Q2 — SHOW WORKING

Write balanced equations for magnesium and aluminium burning separately in excess oxygen. Name both products.

**14** CBSE 2025 · 31/1/1 · Q17 — EXPLAIN

A student says every decomposition must absorb heat. Use decomposition of organic matter in compost to evaluate that claim.

**15** CBSE 2025 · 31/1/1 · Q21 — EXPLAIN

A powder disappears after mixing with water. Why is this observation insufficient to establish a chemical reaction? Name two additional clues.

**16** CBSE 2026 · 31/1/1 · Q17 — SHOW WORKING

Silver nitrate solution is mixed with sodium chloride solution. Write the molecular and net ionic equations.

EXTRA WORKING SPACE

**17** CBSE 2026 · 31/1/1 · Q19 SHOW WORKING

Two formula units of lead nitrate decompose. How many molecules of each gaseous product form according to the equation?

**18** CBSE 2026 · 31/1/1 · Q26 EXPLAIN

In ZnO + C → Zn + CO, name the oxidising agent and justify your answer using oxygen transfer.

**19** CBSE 2026 · 31/1/1 · Q26 SHOW WORKING

Balance the reaction of potassium iodide with lead nitrate and identify the precipitate by name and colour.

**20** CBSE 2026 · 31/1/1 · Q26 EXPLAIN

Compare the energy inputs used to decompose water and silver chloride. Include a balanced equation for each.

EXTRA WORKING SPACE

**21** CBSE 2024 · 31/1/1 · Q27 SHOW WORKING

Mix aqueous sodium sulfate and barium chloride. Write the net ionic equation and name the two spectator ions.

**22** CBSE 2025 · 31/1/1 · Q2 SHOW WORKING

How many moles of oxygen are needed to convert 0.80 mol of aluminium entirely to aluminium oxide?

**23** CBSE 2024 · 31/1/1 · Q21 SHOW WORKING

Using Ca = 40, O = 16 and H = 1, calculate the mass of slaked lime formed from 56 g of pure quicklime with excess water.

**24** CBSE 2023 · 31/1/1 · Q27 SHOW WORKING

If 0.60 mol CO reacts completely with excess $Fe_2O_3$, calculate the amount of Fe formed.

**25** CBSE 2024 · 31/1/1 · Q2 SHOW WORKING

A sample containing 0.30 mol AgCl is fully decomposed by sunlight. Find the amounts of silver and chlorine formed.

EXTRA WORKING SPACE

# 02 Acids, Bases and Salts

**26** CBSE 2023 · 31/1/1 · Q1 — SHOW WORKING

A gas from sodium carbonate and dilute hydrochloric acid turns limewater milky. Write both reactions and name the cloudy solid.

**27** CBSE 2023 · 31/1/1 · Q1 — EXPLAIN

Why does limewater become clear again after a long stream of carbon dioxide passes through it?

**28** CBSE 2023 · 31/1/1 · Q4 — QUICK RESPONSE

Milk of magnesia gives a pH near 10. Classify it and predict what it does to red litmus.

**29** CBSE 2023 · 31/1/1 · Q6 — EXPLAIN

Give the formula of washing soda and distinguish it from anhydrous sodium carbonate.

EXTRA WORKING SPACE

**30** CBSE 2023 · 31/1/1 · Q21 SHOW WORKING

A blue salt loses its colour on heating and turns blue when water is added back. Identify the salt and represent the heating step.

**31** CBSE 2023 · 31/1/1 · Q21 EXPLAIN

How does baking powder differ from baking soda? Which gas causes dough to rise?

**32** CBSE 2023 · 31/1/1 · Q21 SHOW WORKING

Give the equation for heating baking soda. If 0.40 mol decomposes, how much carbon dioxide forms?

**33** CBSE 2023 · 31/1/1 · Q28 EXPLAIN

Milk becomes curd. Predict the direction of its pH change and explain the chemical cause.

EXTRA WORKING SPACE

**34** CBSE 2023 · 31/1/1 · Q28 EXPLAIN

Explain why a basic antacid can help in an example of excess stomach acid. Give one representative equation.

**35** CBSE 2024 · 31/1/1 · Q5 SHOW WORKING

What mass of water is present in 1 mol of washing soda crystals? Use 18 g/mol for water.

**36** CBSE 2024 · 31/1/1 · Q28 EXPLAIN

A pond has pH 7.2 before repeated acid rain and pH 5.3 after. Explain why the change matters for aquatic organisms.

**37** CBSE 2024 · 31/1/1 · Q28 EXPLAIN

Explain the difference between strong and weak acids at the same initial concentration, in terms of ions in water.

EXTRA WORKING SPACE

**38** CBSE 2024 · 31/1/1 · Q28 SHOW WORKING

Two dilute solutions have pH 3 and pH 5. Which is more acidic, and by what factor does their hydronium concentration differ?

**39** CBSE 2025 · 31/1/1 · Q3 QUICK RESPONSE

Arrange pH 13, 5, 7 and 3 in order of increasing acidity.

**40** CBSE 2025 · 31/1/1 · Q4 SHOW WORKING

Complete and balance the neutralisation of magnesium oxide by hydrochloric acid. Explain why the reaction is a neutralisation.

**41** CBSE 2025 · 31/1/1 · Q4 SHOW WORKING

Carbon dioxide is passed into excess sodium hydroxide. Write the equation and classify $CO_2$ by its reaction with the base.

EXTRA WORKING SPACE

**42** CBSE 2026 · 31/1/1 · Q22 QUICK RESPONSE

Match oxalic acid, lactic acid and methanoic acid with one familiar natural source each.

**43** CBSE 2026 · 31/1/1 · Q25 EXPLAIN

Classify $Na_2O$, $CO_2$ and ZnO as basic, acidic or amphoteric. What test distinguishes the amphoteric oxide?

**44** CBSE 2026 · 31/1/1 · Q27a SHOW WORKING

Write the balanced overall equation for electrolysis of brine. State where both gases form.

**45** CBSE 2026 · 31/1/1 · Q27a SHOW WORKING

An ideal chlor-alkali process consumes 0.60 mol NaCl. Calculate the amount of NaOH and $Cl_2$ produced.

EXTRA WORKING SPACE

**46** CBSE 2026 · 31/1/1 · Q27b — EXPLAIN

Name the two starting substances for preparing bleaching powder in the school-level reaction and write the equation.

**47** CBSE 2026 · 31/1/1 · Q27b — SHOW WORKING

Write the heating reaction that converts gypsum to Plaster of Paris without fractional coefficients.

**48** CBSE 2023 · 31/1/1 · Q21 — EXPLAIN

A cast made using Plaster of Paris sets after mixing with water. Explain the chemical change and name the solid formed.

**49** CBSE 2026 · 31/1/1 · Q27b — SHOW WORKING

Complete the preparation of sodium hydrogen carbonate from NaCl, $H_2O$, $CO_2$ and $NH_3$. Identify the other product.

**50** CBSE 2025 · 31/1/1 · Q4 — SHOW WORKING

A sample of sodium hydroxide contains 0.25 mol NaOH. Find the amount of HCl required to neutralise it fully.

EXTRA WORKING SPACE

# 03 Metals and Non-metals

**51** CBSE 2023 · 31/1/1 · Q5 — EXPLAIN

Identify the metal powder commonly used to reduce iron(III) oxide in a thermite mixture. State what kind of oxide that metal forms.

**52** CBSE 2023 · 31/1/1 · Q37 — SHOW WORKING

A copper strip turns black on heating in air. Name the coating and write the balanced equation.

**53** CBSE 2023 · 31/1/1 · Q37 — SHOW WORKING

Demonstrate the amphoteric behaviour of $Al_2O_3$ with one balanced acid reaction and one balanced base reaction.

**54** CBSE 2024 · 31/1/1 · Q1 — SHOW WORKING

Granulated zinc is warmed with aqueous sodium hydroxide. Write the school-level equation and identify the gas.

EXTRA WORKING SPACE

**55** CBSE 2024 · 31/1/1 · Q3 QUICK RESPONSE

Give one metallic element and one non-metallic element that are liquid at about 25 °C.

**56** CBSE 2024 · 31/1/1 · Q17 EXPLAIN

Zinc gives hydrogen with dilute HCl, but generally not with nitric acid. Explain the difference.

**57** CBSE 2024 · 31/1/1 · Q37 EXPLAIN

Set up an electrolytic cell for refining copper. Identify the anode, cathode and electrolyte.

**58** CBSE 2024 · 31/1/1 · Q37 EXPLAIN

During electrolytic refining, why does the pure copper electrode gain mass? Where do insoluble impurities collect?

EXTRA WORKING SPACE

**59** CBSE 2025 · 31/1/1 · Q5 EXPLAIN

An ionic solid does not light a test bulb, but its melt does. Explain in terms of particle motion.

**60** CBSE 2025 · 31/1/1 · Q6 EXPLAIN

Choose an extraction method for magnesium. Explain why carbon reduction is unsuitable.

**61** CBSE 2025 · 31/1/1 · Q7 EXPLAIN

Show the electron transfer when Mg and O form MgO using ionic symbols and electron counts.

**62** CBSE 2025 · 31/1/1 · Q27a EXPLAIN

A zinc carbonate ore is first heated strongly with limited air. Name the process, write its equation and explain its purpose.

EXTRA WORKING SPACE

**63** CBSE 2025 · 31/1/1 · Q27a — SHOW WORKING

A zinc sulfide ore is heated in excess air. Write the balanced equation and name the process.

**64** CBSE 2025 · 31/1/1 · Q28a — EXPLAIN

Compare clean iron nails in dry air, boiled water under oil, and ordinary aerated water. Predict which rusts and why.

**65** CBSE 2026 · 31/1/1 · Q21 — QUICK RESPONSE

Name a solid non-metal that conducts electricity and explain how charge is carried in it.

**66** CBSE 2026 · 31/1/1 · Q23 — EXPLAIN

Calcium initially sinks but then floats when added to water. Explain and give the balanced equation.

EXTRA WORKING SPACE

**67** CBSE 2026 · 31/1/1 · Q28 SHOW WORKING

Write the two reactions used to obtain copper from copper(I) sulfide. Identify the step called self-reduction.

**68** CBSE 2026 · 31/1/1 · Q28 EXPLAIN

What property makes tin–lead solder suitable for joining wires? Explain why the wires need not melt.

**69** CBSE 2026 · 31/1/1 · Q28 SHOW WORKING

In a thermite reaction 0.50 mol $Fe_2O_3$ reacts with excess Al. Find the amount of iron formed and Al used.

**70** CBSE 2026 · 31/1/1 · Q28 SHOW WORKING

In the self-reduction of copper(I) oxide, 0.40 mol $Cu_2O$ reacts with enough $Cu_2S$. Find the amount of copper formed.

EXTRA WORKING SPACE

**71** CBSE 2025 · 31/1/1 · Q5 EXPLAIN

Why does it usually take a high temperature to melt an ionic compound such as magnesium oxide?

**72** CBSE 2025 · 31/1/1 · Q6 EXPLAIN

In molten magnesium chloride electrolysis, name the product at each electrode and write the cathode half-equation.

**73** CBSE 2024 · 31/1/1 · Q37 EXPLAIN

In a copper refining cell the electrodes are accidentally swapped. Why will this fail to refine the original impure copper plate?

**74** CBSE 2025 · 31/1/1 · Q28a EXPLAIN

An iron tool is coated with a continuous layer of paint. Use the conditions needed for rusting to explain the protection.

**75** CBSE 2026 · 31/1/1 · Q28 SHOW WORKING

Calculate the mass of aluminium needed for 1 mol $Fe_2O_3$ in the thermite equation. Take Al = 27 g/mol.

EXTRA WORKING SPACE

# 04 Carbon and Its Compounds

**76** CBSE 2022 · 31/1/1 · Q1 EXPLAIN

Write the molecular and condensed structural formulae of methane and propane.

**77** CBSE 2022 · 31/1/1 · Q1 EXPLAIN

Why does a small molecular carbon compound often boil at a lower temperature than an ionic salt?

**78** CBSE 2022 · 31/1/1 · Q9a DIAGRAM

Draw and name both structural isomers of $C_4H_{10}$. What differs between the molecules?

**79** CBSE 2022 · 31/1/1 · Q9b EXPLAIN

Write the molecular formula of benzene. In its usual school-level alternating-bond sketch, count the C=C bonds and the single bonds including C–H.

EXTRA WORKING SPACE

**80** CBSE 2022 · 31/1/1 · Q9b — SHOW WORKING

Write the general formula of an open-chain alkyne with one triple bond. Determine the molecular formula for five carbons.

**81** CBSE 2023 · 31/1/1 · Q7 — DIAGRAM

Represent cyclohexane and cyclohexene as six-carbon rings. Mark the double bond where relevant and write both molecular formulae.

**82** CBSE 2023 · 31/1/1 · Q34 — SHOW WORKING

An alcohol gives an alkene of molecular mass 28 u on dehydration. Use C = 12 and H = 1 to identify the alkene and the original alcohol.

**83** CBSE 2023 · 31/1/1 · Q34 — EXPLAIN

Ethene is converted to a saturated product by adding hydrogen. Write the reaction and name the catalyst.

EXTRA WORKING SPACE

**84** CBSE 2024 · 31/1/1 · Q4 EXPLAIN

Explain why a liquid made of neutral covalent molecules is usually a poor electrical conductor.

**85** CBSE 2024 · 31/1/1 · Q34a EXPLAIN

Why do consecutive members of the same homologous series show similar reactions but gradually changing boiling points?

**86** CBSE 2024 · 31/1/1 · Q34a SHOW WORKING

Find the difference in relative molecular mass between two consecutive members of a homologous series. Use C = 12 and H = 1.

**87** CBSE 2024 · 31/1/1 · Q34a DIAGRAM

Draw condensed structures for an aldehyde and a ketone both having molecular formula $C_3H_6O$.

EXTRA WORKING SPACE

**88** CBSE 2024 · 31/1/1 · Q34b — SHOW WORKING

Give the dehydration equation for ethanol and state the temperature and reagent.

**89** CBSE 2025 · 31/1/1 · Q34a — SHOW WORKING

Balance the complete combustion of ethanol. How many moles of oxygen are used per mole of ethanol?

**90** CBSE 2025 · 31/1/1 · Q34a — SHOW WORKING

Give the equation for hydrogenating propene. What mass of hydrogen is needed for 0.50 mol propene? Use $H_2$ = 2 g/mol.

**91** CBSE 2025 · 31/1/1 · Q34b — EXPLAIN

A piece of sodium reacts with dry ethanol. Name both products and write the equation.

EXTRA WORKING SPACE

**92** CBSE 2025 · 31/1/1 · Q34b — SHOW WORKING

If 0.30 mol of ethanol reacts with excess sodium, calculate the amount of hydrogen released.

**93** CBSE 2026 · 31/1/1 · Q20 — QUICK RESPONSE

Write the functional group of aldehydes and locate it in $CH_3CHO$.

**94** CBSE 2026 · 31/1/1 · Q24 — EXPLAIN

Explain why carbon generally shares its four valence electrons in forming stable compounds.

**95** CBSE 2026 · 31/1/1 · Q29a — EXPLAIN

Why does ordinary soap form scum in hard water? Which ions are responsible?

EXTRA WORKING SPACE

**96** CBSE 2026 · 31/1/1 · Q29a — EXPLAIN

Compare the ability of carbon and silicon to form long chains. Relate the difference to bond strength.

**97** CBSE 2026 · 31/1/1 · Q29a — SHOW WORKING

Show how ethanol is converted to ethanoic acid by oxidation. Name one suitable oxidising reagent.

**98** CBSE 2026 · 31/1/1 · Q29b — SHOW WORKING

A sweet-smelling liquid is produced by warming ethanoic acid with ethanol. Write the equation and name the reaction.

**99** CBSE 2026 · 31/1/1 · Q29b — SHOW WORKING

Write the reaction of ethyl ethanoate with aqueous sodium hydroxide. Name the reaction type and products.

**100** CBSE 2026 · 31/1/1 · Q29b — SHOW WORKING

A batch starts with 0.20 mol ethanoic acid and excess ethanol. What is the maximum mass of ethyl ethanoate? Use 88 g/mol.

EXTRA WORKING SPACE

# The reference shelf

Each citation identifies the source idea in a CBSE Class 10 Science board paper. Prompts are paraphrased, split into individual skills, or extended with fresh data. Question labels refer to the source paper, and worksheet numbers refer to this assignment.

**2022 · Science 086 · Paper 31/1/1 · Term II**

Open question paper reference

**2023 · Science 086 · Paper 31/1/1**

Open question paper reference

**2024 · Science 086 · Paper 31/1/1**

Open question paper reference

**2025 · Science 086 · Paper 31/1/1**

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